Swarm Witness: What Do Agent-Safety Mechanisms Actually Establish?

Tracks: 3 (Strengthening Agent Infrastructure), 4 (Multi-Agent Oversight and Control)

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Abstract

After matching policy, observations, trust assumptions, durability, and enforcement coverage, which claimed benefits of agent-safety mechanisms remain? We test that question with Swarm Witness, an executable claim-audit kit that pins third-party artifacts, states precise claim cards, runs discriminating checks, and records fail-closed conclusion labels. Classical reductions (recover-before-admit; no-implicit-exclusivity; durable grant-table checks) are questions under test via matched baselines—not intro verdicts. On three pinned targets we obtain: SoundGate recovery-barrier (R1) reproduced — 153 histories/policy, original prohibited effects 143 vs barrier 0, with 147 strict refusal-startup refusals; EvidenceGraph receipt stress (R2) reproduced — 36/36 condition-by-seed checks, old-rule relay wrong attributions [1,1,1] per seed vs safer supported [0,0,0], and clean withheld-without-exclusivity [12,12,12]; BMO retained-grant vs ClassicalGate (R3 decision-model) unresolved — matched decision-model comparison incomplete (0/4 isomorphic BMO observations; selected Go package tests are metadata only); separately, derived protocol r3-ledger-claim-v1 reports a bounded L1 ledger-claim match (`classical_baseline_matches`) for six synthetic cases under native pin `4b717d7` (dispositions and property-relevant projections at every step; not full identity-hash equality or general BMO equivalence). Weekend contributions claimed here are (1) pinned replications for R1/R2, (2) the audit kit, and (3) a scoped L1 ledger-claim comparison distinct from the unresolved decision-model R3. Auditors should publish matched baselines and labels.

1. Introduction

Research question. After matching policy, observations, trust assumptions, durability, and enforcement coverage, which claimed benefits of agent-safety mechanisms remain?

Authorization gates, receipt graphs, and retained grants are agent-facing controls that auditors must score carefully. When a report claims a restart-safe refusal, receipt-based attribution, or retained-grant relaunch control, the open question is whether the result is a new property, a repaired instance of a classical mechanism, or an incomplete comparison. Classical reductions — recover-before-admit for durable logs, no-implicit-exclusivity for receipt tokens, and durable grant-table checks before admit—are **questions we test** with matched baselines, not conclusions we assert in the introduction.

Threat / failure modes. (i) An admission gate forgets an acknowledged rejection across restart when the durable log is not recovered before admit. (ii) A verifier treats possession of a copied receipt token as event attribution without exclusivity or authenticity. (iii) Retained start/grant metadata is treated as standing relaunch authority without consulting a durable grant table.

Main contributions (claimed).

1. **Audit kit:** Swarm Witness pins artifacts, types Claim/Run/Finding cards, states classical baselines, runs discriminating experiments, and fails closed on evidence consistency (`reproduced`, `unresolved`, …).
2. **Pinned replications:** SoundGate recovery barrier (R1) and EvidenceGraph receipt stress (R2) under published harnesses, with metrics from committed `evidence/*/summary.json`.

A broad methodological finding that BMO's durable-grant story reduces to classical controls is **not claimed**: the legacy decision-model R3 comparison remains incomplete (`unresolved`). Separately, derived protocol **r3-ledger-claim-v1** contributes a **bounded L1 ledger-claim match** under pin `4b717d7` (see §4.4), not a general equivalence claim.

Tracks: **3** (Strengthening Agent Infrastructure) and **4** (Multi-Agent Oversight and Control). Preexisting BMO code and Oct 25 scaffold are marked in `PREEXISTING.md`; judged weekend work is the audit method, pins, baselines, experiments, and evidence packaging.

2. Related Work

We build on Apart Swarm Alignment themes of identity, durable commitments, and oversight under copy/rename/relay (Hammond et al., 2025; Chan et al., 2025; Greenblatt et al., 2024). Concrete targets:

- **SoundGate / "A Restart Is Not Permission"** — durable auth decisions and recovery-barrier repair (pinned package `soundgate==0.1.3`, source revision `6568403`).
- **EvidenceGraph / Junejo "Receipt Possession Is Not Event Attribution"** — stress harness at git rev `a5ea5c9`; paper baseline rev `efbd419` (pre-extension behavior; no runnable stress script at that rev).
- **BMO ordinary-retry retained grant** — preexisting durable grant / owner-witness tests under pinned `bmo_root` / `git_rev` (read-only `go test` as metadata).

In-repo baseline cards state mechanism classes without equating every proxy to a decision match. Swarm Witness differs from building a new gate: it asks what labels survive matched baselines and discriminating checks.

3. Methods

Claim cards (`claims/*.yaml`) fix `property`, `scope`, `quantifier`, `success_criterion`, and `assumptions` (plus source_ref/version and original vs operational interpretation). Classical baselines live in `baselines/*.md` as authored notes, not Claim fields. **Pins** (`artifacts/pins.yaml`) freeze revisions and hashes. **Baselines** state the mechanism class without equating every proxy to a decision match. **Evidence** uses Finding labels; `scripts/check_evidence.py` fails closed if `evidence/index.yaml` disagrees with findings or supporting refs are missing. The checker's metric-binding scope is limited: reported metrics must agree with linked Run observations when keys overlap—it does not re-execute harnesses.

R1 (SoundGate). Rebuild and sweep the paper recovery-barrier harness at pinned source `6568403`. Compare policies `original`, `strict`, and `barrier` over the published histories. Success for reproduction: barrier `prohibited_effects == 0` with original showing the defect. Full `raw.json` is gitignored; committed `evidence/r1/raw.json.gz` supports local inspection (`gzip -dc … > raw.json`). Recorded `raw_sha256` is integrity only—hash alone does not replace inspecting analysis case records. Supplemental cargo `wal_recovery_barrier` on 0.1.4 recorded in notes.

R2 (EvidenceGraph). The harness **runs at `a5ea5c9`**, which publishes `incident_stress.py`. Paper baseline `efbd419` is the pre-extension tree (no stress script); baseline *behavior* is represented by replaying the authors' later-published stress pack at `a5ea5c9` (old-rule vs safer-default contrast), not by executing a missing harness at `efbd419`. Label `reproduced` when all contract checks pass.

R3 (two distinct protocols). Legacy **r3-matched-decision-model** uses shared `cases.yaml` decision-model inputs; ClassicalGate emits allow/deny per case. BMO decisions require isomorphic observations—never package pass counts. When any required BMO case is unmatched, that comparison Finding is **`unresolved`**. Derived **r3-ledger-claim-v1** separately compares ClassicalLedgerGate (SQLite) to native `ClaimOrdinaryRetainedGrantFirstEditTx` under pin `4b717d7` with fail-closed native qualification.

4. Results

Labels from `evidence/index.yaml` / findings only: R1 **reproduced**, R2 **reproduced**, R3 decision-model **unresolved**, R3 ledger-claim-v1 **classical_baseline_matches** (bounded L1). Metrics cite `evidence/*/summary.json`.

4.1 R1 — SoundGate recovery barrier (`reproduced`)

Fresh local rebuild+sweep; `all_checks_pass: true`; `raw_sha256=51b4acd6…`. Histories per policy: **153**.

Policy	cases	prohibited_effect	refused_before_ack	supported	fresh_work
original	153	**143**	4	6	149
strict	153	0	**147**	6	6
barrier	153	**0**	0	**153**	153

Barrier restores acknowledge-and-restrict across restart under the published histories; original loses restrictions (143 prohibited effects). Cargo `wal_recovery_barrier` on 0.1.4: 3 passed (notes). Classical recover-before-admit is an authored baseline interpretation (`baselines/r1_wal_recover_before_admit.md`), not a separate empirical `classical_baseline_matches` Finding.

4.2 R2 — Receipt ≠ attribution (`reproduced`)

Harness at `a5ea5c9`; `condition_by_seed_checks` **36**, passed **36**; seeds 0,1,2; `all_contract_checks_pass: true`; `source_tree_sha256=2962568f…`.

Metric (per seed)	seed 0	seed 1	seed 2
old_rule wrong attributions on relay	1	1	1
safer supported on relay	0	0	0
safer correct on relay	0	0	0
clean withheld without exclusivity	12	12	12
shared_tokens correct	0	0	0
false_exclusivity confident errors	1	1	1

Relay seed-0 outcomes: ambiguous 10, unmatched 2. Safer default withholds attribution without exclusivity; old rule errs once per seed on relay. Classical no-implicit-exclusivity is authored interpretation (`baselines/r2_no_implicit_exclusivity.md`).

4.3 R3 — Retained grant vs ClassicalGate (`unresolved`)

Matched **decision-model** comparison: **0/4** isomorphic BMO observations; `missing_bmo_cases: 4`. Comparison Finding `r3-f-comparison` label **`unresolved`** — comparison not performed / incomplete. ClassicalGate arm completed all four cases under the decision model (`state`, `request`, `grant_version`); BMO arm records `execution_status=missing` / `missing_isomorphic_observation` for each. Selected preferred Go tests on the pinned BMO revision remain available as package metadata and are **not** used as decisions or as a substitute comparative label.

Case	ClassicalGate	BMO decision	BMO status
retained_start_claim_relaunch	deny	—	missing
rejected_execute	deny	—	missing
active_grant_execute	allow	—	missing
stale_grant_version	deny	—	missing

4.4 R3 — Ledger claim L1 (`classical_baseline_matches`)

Derived protocol **r3-ledger-claim-v1** is distinct from §4.3. Across six synthetic cases (T01–T06), ClassicalLedgerGate and the native BMO L1 bridge at pin `4b717d7` recorded equal dispositions and property-relevant projections (grant_spent, counts, call_id sets) at every step, and both arms satisfied the declared one-use / reopen / rollback properties. Finding `r3-ledger-claim-v1-comparison` label **`classical_baseline_matches`** under fail-closed native qualification (manifest, exit 0, adapter identity, git-archive pin inventory, bridge hashes). Cross-arm equality does **not** claim native grant/epoch hash identity parity with synthetic classical aliases; durable identity rows are archived under `observed_identities` for audit. This does not establish event-by-event observational equivalence beyond the declared projection, live execution safety, process-restart correctness, concurrency, or general BMO equivalence.

5. Discussion

Implications. For Tracks 3–4, durable identity and oversight claims should be scored against classical admit and evidence rules under matched observations. R1 and R2 show that published repairs can **reproduce** under pinned harnesses; that is valuable engineering and audit evidence, not proof of a new multi-agent primitive. Legacy R3 shows the fail-closed path: without isomorphic decision-model BMO observations, that comparative audit stays **unresolved**. The separate L1 ledger-claim protocol shows a bounded match is possible when both arms emit complete native observations under a pinned bridge.

Limitations (summary). ClassicalGate is a **decision-model** null (not restart durability); legacy R3 comparison is incomplete; ledger-claim equality is projection-scoped (not full identity-hash equality); R1 raw inspectability is via committed `raw.json.gz`; R2 executable binding is `a5ea5c9` with `efbd419` as behavioral baseline via published stress pack; evidence checker metric-binding is scope-limited. Full Limitations and Dual-Use appendix follows.

Future work. Shared observation APIs for decision-model R3 isomorphic matches; optional explicit grant/epoch alias map with negative controls; expand pins; automate baseline-vs-claim diffs for continuous oversight.

Appendix A. Limitations and Dual-Use / Ethical Considerations

What each result does and does not establish

R1 (`reproduced`). Establishes that, on the pinned SoundGate recovery-barrier harness (source revision `6568403`), the barrier policy yields `prohibited_effects = 0` across 153 histories while the original policy yields 143 prohibited effects, with 147 strict refusal-startup refusals under the strict policy. Does **not** establish that every agent admission gate is restart-safe, that SoundGate is free of other defect classes, or that recover-before-admit is a new multi-agent invention (that classical reduction remains an authored baseline question under test). Committed `evidence/r1/raw.json.gz` supports local inspection of the sweep artifact (`gzip -dc evidence/r1/raw.json.gz > evidence/r1/raw.json`); recorded `raw_sha256` is an integrity check and **does not replace** inspecting `raw.json` / analysis case records.

R2 (`reproduced`). Establishes that the stress pack executed at harness rev `a5ea5c9` passes 36/36 condition-by-seed checks: old-rule relay wrong attributions are 1 per seed; safer relay supported attributions are 0; clean runs withhold all 12 attributions without exclusivity declarations. Paper baseline `efbd419` is the pre-extension tree (no `incident_stress.py`); its baseline *behavior* is represented by replaying the authors' later-published stress pack at `a5ea5c9`, not by running a missing harness at `efbd419`. Does **not** establish that all receipt or ledger systems are safe under relay/copy, or that exclusivity declarations themselves cannot be gamed outside the modeled threat surface.

R3 decision-model (`unresolved`). Establishes that the matched ClassicalGate decision-model comparison against BMO is **incomplete**: 0/4 required cases have isomorphic BMO observations, so the comparison Finding is `unresolved` (comparison not performed). ClassicalGate is a **decision-model** null (`state`, `request`, `grant_version`)—not a restart-durability claim. Selected preferred Go tests on the pinned BMO revision are package metadata only and do not fill decisions or substitute for a matched comparative label. Does **not** establish ClassicalGate ↔ BMO API equivalence, assumption-dependent affinity, or a methodological reduction of BMO's durable-grant story to classical controls.

R3 ledger-claim-v1 (`classical_baseline_matches`, bounded). Establishes equal dispositions and property-relevant projections (grant_spent, counts, call_id sets) across T01–T06 for ClassicalLedgerGate vs native `ClaimOrdinaryRetainedGrantFirstEditTx` at pin `4b717d7` under fail-closed native qualification. Does **not** claim synthetic classical grant/epoch aliases equal native durable hashes, full event-trace observational equivalence, live execution safety, process-restart correctness, concurrency behavior, or general BMO equivalence. Legacy decision-model R3 remains separately unresolved.

Finite experiments and checker scope

Pinned histories, seeds `{0,1,2}`, and preferred Go tests are finite. Passing them does not imply equivalence across versions, storage backends, concurrent writers outside the model, or production deployments. Labels apply to the stated scope on the claim cards only. `scripts/check_evidence.py` binds reported metrics to linked Run observations when keys overlap; it does not re-execute harnesses or prove metric completeness beyond that binding.

Dual-use / adversarial use of the audit method

An executable claim-audit method can help **defenders** find weak gates and attribution gaps—and can also help **adversaries** locate the same gaps (restart races, possession-as-attribution, retained-metadata relaunch). We publish **tests, pins, matched baselines, and labels**, not

exploit kits, weaponized harnesses, or production credentials. Reproduction artifacts are scoped to already-public or author-local preexisting trees (`PREEXISTING.md`).

Preexisting code vs weekend analysis

Oct 25 scaffold paths and the external BMO tree at pinned `bmo_root` / `git_rev` predate this sprint and were not modified for R3 beyond read-only `go test`. Weekend work is the Swarm Witness schema, pins, claim cards, baselines, experiment wrappers, evidence records, index checker, and this report. Judges should score the audit contribution, not treat preexisting BMO surfaces as sprint inventions.